1.5 Cellular Respiration

Unit 1: Cellular Functions

Overview

> ■ **Key Concept**
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> Cellular respiration is the process by which cells break down glucose to release energy in the form of ATP. Overall equation: C■H■■O■ + 6O■ → 6CO■ + 6H■O + ATP (energy)

This is an aerobic (oxygen-requiring) process that occurs primarily in the mitochondria. It proceeds in three main stages: Glycolysis, the Krebs Cycle, and the Electron Transport Chain (Oxidative Phosphorylation).

Stage 1: Glycolysis

Glycolysis splits glucose into two 3-carbon pyruvate molecules. It produces a small net gain of 2 ATP and 2 NADH (electron carriers). This step occurs in ALL organisms.

Term	Definition
Location	Cytoplasm (cytosol)
Input	1 glucose (6C)
Output	2 pyruvate (3C each), net 2 ATP, 2 NADH
Oxygen required?	No — anaerobic step

Pyruvate Oxidation (Bridge Reaction)

Each pyruvate enters the mitochondrial matrix and is converted to Acetyl-CoA (2C) + CO■, producing 1 NADH per pyruvate (2 total per glucose). This prepares pyruvate for the Krebs cycle.

Stage 2: Krebs Cycle (Citric Acid Cycle)

Acetyl-CoA combines with oxaloacetate to form citrate (6C). Through a series of reactions, CO_2 is released and electron carriers (NADH, $FADH_2$) are produced. Oxaloacetate is regenerated to accept another Acetyl-CoA.

Term	Definition
Location	Mitochondrial matrix
Input	2 Acetyl-CoA (one cycle per pyruvate → 2 cycles per glucose)
Output per 2 cycles	4 CO_2, 2 ATP, 6 NADH, 2 $FADH_2$

Stage 3: Electron Transport Chain + Oxidative Phosphorylation

> ■ **Tip**
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> This is where the vast majority of ATP is produced — ~32-34 out of ~36-38 total ATP per glucose molecule.

Electrons from NADH and $FADH_2$ are passed along protein complexes in the inner mitochondrial membrane. This releases energy used to pump H^+ ions across the membrane, creating a gradient. H^+ flows back through ATP synthase, driving ATP production. O_2 is the final electron acceptor, combining with H^+ to form water.

Term	Definition
Location	Inner mitochondrial membrane (cristae)
Input	NADH, $FADH_2$ (electron carriers from previous stages)
Output	~32–34 ATP, H_2O

Anaerobic Respiration (Fermentation)

When oxygen is absent, cells cannot run the ETC. Fermentation regenerates NAD^+ so glycolysis can continue, producing a small amount of ATP.

Lactic acid fermentation (animals, some bacteria): pyruvate → lactic acid. Causes muscle fatigue during intense exercise.

Alcoholic fermentation (yeast, some plants): pyruvate → ethanol + CO_2. Used in bread-making and brewing.